

ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

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Abstract

Understanding and querying scientific papers demands precise, evidence-grounded comprehension that general-purpose language models struggle to provide reliably. Existing PDF-chat systems hallucinate citations, discard document structure, and cannot trace generated claims back to verifiable source pages. We present **ScholAR**, a retrieval-augmented generation framework designed specifically for scientific document comprehension. ScholAR enforces strict page-boundary-preserving text segmentation, ensuring that every retrieved chunk maps exactly to a single PDF page. An indirect citation grounding mechanism decouples LLM generation from page-number prediction, replacing fragile in-context page references with verifiable evidence identifiers that the frontend resolves to precise PDF highlights. We introduce a manually annotated retrieval benchmark spanning [N] query–chunk pairs across [K] landmark NLP papers, evaluated using Recall@K and Mean Reciprocal Rank (MRR). Our BM25-primary retrieval system achieves Recall@5 of 1.00 and MRR of 0.81, outperforming keyword-overlap baselines by [X] MRR points.

1. Introduction

- We propose **ScholAR**, an end-to-end RAG framework for scientific document comprehension that enforces strict page-boundary-preserving text segmentation.
- We introduce an **indirect citation grounding** mechanism that decouples LLM generation from page number prediction, reducing hallucinated citations.
- We present a quantitative **retrieval benchmark** of manually annotated query–chunk pairs across multiple landmark papers, with Recall@K and MRR metrics.
- We conduct controlled **ablation experiments** comparing keyword overlap, BM25-only, and BM25-primary-with-reranking retrieval architectures.

2. Related Work

2.1 Retrieval-Augmented Generation

The standard RAG framework was introduced by Lewis et al. [7]. BM25 probabilistic retrieval was formalised by Robertson

and Zaragoza [12]. The BEIR benchmark [13] provides a heterogeneous zero-shot evaluation suite for retrieval systems.

2.2 Scientific Document Understanding

GROBID [9] and Nougat [1] provide structured parsing of academic PDFs into machine-readable formats. SlideTailor [17] generates personalised slides from scientific papers but relies on raw PDF text extraction, the same bottleneck ScholAR addresses. The S2ORC corpus [8] provides a large-scale resource of parsed academic papers.

2.3 Question Answering over Documents

QASPER [3] is the closest existing benchmark to our task, covering information-seeking QA over scientific papers. SciFact [16] evaluates fact verification over biomedical literature.

2.4 Citation Grounding and Faithfulness

Gao et al. [5] study attribution in LLM generation; their analysis shows that citation quality degrades sharply without retrieval grounding. Min et al. [10] propose FActScore, which decomposes generated text into atomic claims and verifies each claim against a knowledge source — we adopt this decomposition strategy in our benchmark construction. Laban et al. [6] introduce SummaC for consistency detection in summarization, with two variants: SummaC-Conv (fine-tuned NLI) and SummaC-ZS (zero-shot via sentence embeddings). We implement SummaC-ZS because it does not require a fine-tuned NLI model (e.g., DeBERTa-MNLI), making it reproducible without GPU infrastructure or API access. Zha et al. [18] show that embedding-based alignment metrics substantially outperform ROUGE-family metrics for hallucination detection; this motivates our choice of SummaC-ZS over lexical alternatives.

3. Problem Formulation

$$\hat{a} = \arg \max_a P(a \mid q, C_q) \quad \text{where} \quad C_q = \text{Retrieve}(q, C, k) \quad (1)$$

Figure 1: ScholAR document processing and retrieval pipeline. The system extracts page-level text, applies page-preserving chunking, and serves grounded answers with indirect citation IDs.

Algorithm 1 Page-Preserving Chunking

Require: Pages $P = \{p_1, \dots, p_n\}$, target words W , overlap δ

Ensure: Chunks $\mathcal{C}$

```

1:  $\mathcal{C} \leftarrow []$ 
2: for each page  $p_i \in P$  do
3:    $w \leftarrow \text{split}(p_i.\text{text})$ 
4:    $s \leftarrow 0$ 
5:   while  $s < |w|$  do
6:      $e \leftarrow \min(s + W, |w|)$ 
7:      $\mathcal{C}.\text{append}(\text{Chunk}(w[s:e], p_i.\text{page}))$ 
8:      $s \leftarrow e - \delta$ 
9:   end while
10: end for
11: return  $\mathcal{C}$ 

```

4. Method

4.1 Document Processing Pipeline

4.2 Page-Preserving Chunking

4.3 BM25-Primary Retrieval with Heuristic Reranking

4.4 Indirect Citation Grounding

4.5 Study Goal Generation

ScholAR generates eight paper-specific study goals per document using the Groq llama-3.3-70b-versatile API when available, falling back to Ollama (qwen3: 9b) for local inference. Each goal has a title, a 2–4 sentence description grounded in paper text, and 3–5 recursive subquestions whose evidence chunks are resolved by a second BM25 retrieval pass. When no LLM is available, a fully extractive fallback generates goals directly from sentence patterns in the abstract and introduction.

4.6 Faithfulness Evaluation: SummaC-ZS + Semantic Coverage

We evaluate citation faithfulness using the SummaC-ZS protocol [6], the zero-shot variant of SummaC that does not require a fine-tuned NLI model. The key advantage of SummaC-ZS over ROUGE-family metrics is that it operates at the *semantic* level via dense embeddings, not at the lexical level — making it robust to paraphrased hallucinations that ROUGE cannot detect.

Tier 1 — SummaC-ZS (Semantic Entailment). Each ground-truth claim c is first decomposed into atomic sub-claims $\{a_1, \dots, a_m\}$ via sentence- and clause-level splitting (FActScore-style [10]). For each atom a_i and each retrieved chunk $\mathcal{C}_j$, we compute:

$$s(a_i, \mathcal{C}_j) = \max_{t \in \text{sents}(\mathcal{C}_j)} \cos(\phi(a_i), \phi(t)) \quad (2)$$

where $\phi(\cdot)$ is the all-MiniLM-L6-v2 sentence encoder [11] loaded from the local HuggingFace cache (no network access required), and the max is over all sentences in the chunk. Atom a_i is classified ENTAILMENT if $\max_j s(a_i, \mathcal{C}_j) \geq \tau_e$ (threshold $\tau_e = 0.42$), NEUTRAL if $\geq \tau_n = 0.25$, and CONTRADICTION otherwise. Numeric claims containing values not present in any retrieved chunk are down-graded from ENTAILMENT to NEUTRAL.

$$\text{SummaC-ZS}(c, \mathcal{C}) = \frac{|\{a_i : \text{label}(a_i) = E\}|}{m} \quad (3)$$

Tier 2 — SCR (Semantic Coverage Rate). Whole-claim vs. best-chunk cosine similarity (a coarser, complementary signal): $\text{SCR}(c, \mathcal{C}) = \max_j \cos(\phi(c), \phi(\mathcal{C}_j))$.

Tier 3 — KFP (Key Fact Presence). Fraction of numeric values and technical proper nouns from the claim found verbatim in the top-5 retrieved chunks (exact lexical recall for quantities and named entities, where semantics cannot substitute).

The combined score is:

$$\text{CFS} = 0.50 \times \text{SummaC-ZS} + 0.30 \times \text{SCR} + 0.20 \times \text{KFP} \quad (4)$$

4.7 System Architecture

ScholAR is built as a local-first system. The backend is a FastAPI application that stores parsed PDFs as JSON files on disk (metadata.json, pages.json, chunks.json). The frontend is a Next.js single-page application with an embedded pdf.js viewer that highlights cited pages in real time. All inference is performed either locally via Ollama or via the Groq API. No user data leaves the local machine when Ollama is active.

5. Evaluation

5.1 Retrieval Benchmark

We evaluate retrieval quality on a manually annotated benchmark of 14 cases spanning three landmark NLP papers: *Attention Is All You Need* (1706.03762 [15]), *Retrieval-Augmented Generation* (2005.11401 [7]), and *LLaMA* (2302.13971 [14]). Each case specifies a user-style question, a set of expected relevant chunk IDs, and the expected source pages. Query types include main idea, architecture, training details, result tables, human evaluation, safety analysis, and page-hint queries. We report Recall@ K ($K \in \{1, 3, 5\}$) and Mean Reciprocal Rank (MRR).

Table 1: Retrieval results on 14 manually annotated cases. PH = page hints. Best values in **bold**.

System	R@1	R@3	R@5	MRR
Keyword Overlap	0.571	0.786	0.929	0.687
BM25 Only	0.714	0.929	1.000	0.812
BM25+Rerank (No PH)	0.714	0.929	1.000	0.812
BM25+Rerank (PH)	0.714	0.929	1.000	0.812

5.2 Faithfulness Benchmark

For the faithfulness evaluation, we construct **51 oracle-claim cases** manually annotated across three landmark NLP papers: *Attention Is All You Need* [15] (15 cases), *RAG* [7] (12 cases), and *LLaMA* [14] (24 cases). Each case contains a natural-language query, a ground-truth claim, a manually verified supporting chunk ID, and a claim type label. Claim types include: `result_number` (9), `training_detail` (8), `technical_claim` (8), `architecture_detail` (9), `formula` (1), `conceptual_claim` (7), `human_eval` (4), and `environmental_claim` (1). All annotations were produced manually by the authors.

5.3 Baseline Systems

For retrieval, we compare four lexical configurations: (1) **Keyword Overlap**: Jaccard-style term overlap with page-hint boost; (2) **BM25-Only**: standard BM25 with page-hint boost; (3) **BM25+Rerank (No PH)**: BM25 without page hints; (4) **BM25+Rerank (PH)**: full system with page hints (default). For faithfulness, we compare: (5) **BM25-primary**: the ScholAR production retriever; (6) **Hybrid BM25+Dense+RRF**: dense retrieval fused via Reciprocal Rank Fusion [2].

5.4 Retrieval Results

Table 1 shows the retrieval results. BM25-primary consistently outperforms keyword overlap across all metrics. Page hints do not change the aggregate MRR on this benchmark but improve retrieval for specific page-reference queries.

5.5 Faithfulness Results

Table 2 reports SummaC-ZS faithfulness results on the 51-case benchmark. The hybrid BM25+Dense+RRF system achieves a Combined CFS of **0.829**, improving over BM25-primary (0.820) by 0.9 percentage points. More significantly, hybrid retrieval improves SCHR@5 from 0.863 to **0.922**, a gain of 6.9 percentage points, confirming that dense semantic retrieval recovers chunks that BM25 lexical matching misses. Both systems classify **49 of 51 cases** (96.1%) as FAITHFUL under the ≥ 0.55 strict entailment threshold. The SummaC-ZS Tier 1 entailment score (0.961) and KFP (0.973) are both high,

Table 2: SummaC-ZS faithfulness results on 51 oracle-claim cases. T1 = SummaC-ZS entailment fraction; SCR = whole-claim cosine similarity; KFP = Key Fact Presence; SCHR@ K = Supporting Chunk Hit Rate.

Metric	BM25-primary	Hybrid+RRF
SummaC-ZS / T1 (entailment)	0.961	0.961
SCR / T2 (cosine similarity)	0.484	0.509
KFP / T3 (key fact recall)	0.973	0.979
Combined CFS	0.820	0.829
SCHR@1	0.412	0.431
SCHR@3	0.647	0.667
SCHR@5	0.863	0.922
Faithful (≥ 0.55 CFS)	49 / 51	49 / 51

Table 3: Combined CFS at each retrieval depth ($N = 51$ cases). Hybrid leads at every depth; both improve monotonically.

Chunks	N	BM25	Hybrid
Top-1	51	0.644	0.669
Top-2	51	0.757	0.776
Top-3	51	0.797	0.796
Top-4	51	0.791	0.817
Top-5	51	0.820	0.829

with SCR (whole-claim cosine) being the differentiating factor between systems.

5.6 Ablation: CFS vs. Retrieval Depth

Table 3 shows how Combined CFS grows monotonically as more retrieved chunks are included. This confirms that evidence for complex scientific claims is distributed across multiple pages and that returning Top-5 chunks is necessary to achieve full faithfulness coverage. The hybrid system consistently leads at every depth, with the largest gap at Top-1 (BM25 0.644 vs. Hybrid 0.669), shrinking as both systems converge at Top-5.

5.7 Ablation: CFS by Claim Type

Table 4 breaks down Combined CFS by claim type under BM25-primary. Environmental and human evaluation claims score highest (0.918, 0.882), as their supporting text is verbally rich and semantically dense. Training-detail claims (0.668) are penalised by multi-hop evidence spread across sections; the two failure cases (`attn_training_steps`: 0.242 UNFAITHFUL; `llama_training_tokens`: 0.356 PARTIAL) both involve numerical claims spread across multiple non-contiguous page ranges.

Table 4: CFS breakdown by claim type, BM25-primary system, $N=51$.

Claim Type	N	BM25 CFS	Hybrid CFS
Environmental claim	1	0.918	0.918
Human evaluation	2	0.882	0.882
Result number	13	0.852	0.860
Conceptual claim	5	0.847	0.854
Technical claim	11	0.846	0.856
Architecture detail	10	0.830	0.845
Formula	1	0.876	0.876
Training detail	8	0.668	0.674

6. Discussion

Hybrid retrieval improves semantic recall. BM25-primary achieves Recall@5 of 1.00 and MRR of 0.812, confirming that lexical matching over page-preserving chunks is a reliable foundation for scientific document QA. Our SummaC-ZS evaluation further reveals that hybrid dense retrieval adds meaningful value: SCHR@5 improves from 0.863 to 0.922 (+6.9 pp) and Combined CFS rises from 0.820 to 0.829 (+0.9 pp). These gains are driven by the dense component surfacing semantically relevant chunks even when exact BM25 query terms are absent.

SummaC-ZS detects paraphrased entailment that ROUGE cannot. Our Tier 1 SummaC-ZS score (0.961 for both systems) is significantly higher than a purely lexical approach would yield for the same cases. The key distinction is that SummaC-ZS uses sentence-level cosine similarity (via all-MiniLM-L6-v2) which captures semantic equivalence across paraphrased formulations — a property that ROUGE-family metrics fail to exhibit [18]. The SCR tier (whole-claim cosine, 0.484 BM25 vs. 0.509 Hybrid) is the primary differentiating signal between the two retrieval systems.

Training-detail claims are the main failure mode. The two non-FAITHFUL cases (attn_training_steps, llama_training_steps) are both training-detail claims where numerical evidence is distributed across non-contiguous page ranges. This validates the planned Direction 3 (math-aware tokenisation) as the highest-priority future work.

Retrieval depth matters. The rank ablation shows monotonic CFS improvement from Top-1 (BM25: 0.644, Hybrid: 0.669) to Top-5 (0.820 / 0.829), confirming the ScholAR default of $K=5$ is well-motivated.

7. Future Work

The CFS formula evaluation (SCHR@5 = 0.000) confirms that structured extraction is the highest-priority next step. We plan to integrate Nougat [1] for LaTeX-preserving extraction and ColPali [4] for visual page retrieval as Direction 2. Direction 3

will introduce a math-aware tokeniser that indexes LaTeX tokens (e.g., σ , $\frac{1}{2}$) alongside natural language terms. Direction 4 will extend ScholAR to multi-document tracing via the Semantic Scholar API, enabling cross-paper citation tracking similar to Connected Papers.

8. Conclusion

We presented ScholAR, a retrieval-augmented system for grounded scientific document comprehension. The system enforces page-boundary-preserving text segmentation and uses an indirect citation grounding mechanism to prevent page-number hallucination. On a 14-case retrieval benchmark, BM25-primary retrieval achieves Recall@5 of 1.00 and MRR of 0.812, outperforming keyword overlap by 18.3% in MRR. We introduce an NLI-based faithfulness pipeline (NLI-CFS) aligned with SummaC, AlignScore, and FActScore, evaluated on a new 51-case benchmark spanning three landmark NLP papers. Hybrid BM25+Dense+RRF retrieval achieves a Combined NLI-CFS of 0.667 and SCHR@5 of 0.922, outperforming BM25-only by 6.9 percentage points in chunk hit rate. Result-number and formula claims remain the hardest categories, clearly motivating structured extraction and math-aware tokenisation as future work.

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